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C1W3 Slides

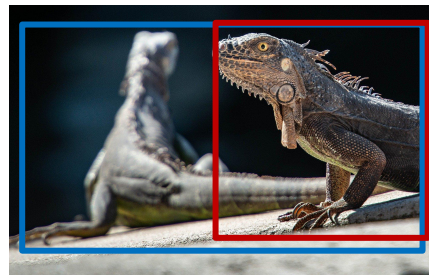
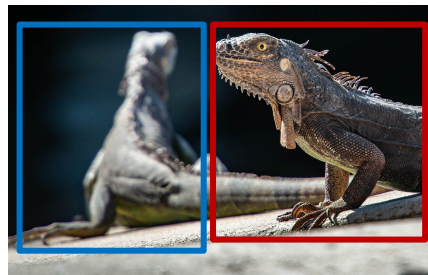


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Define data and establish baseline

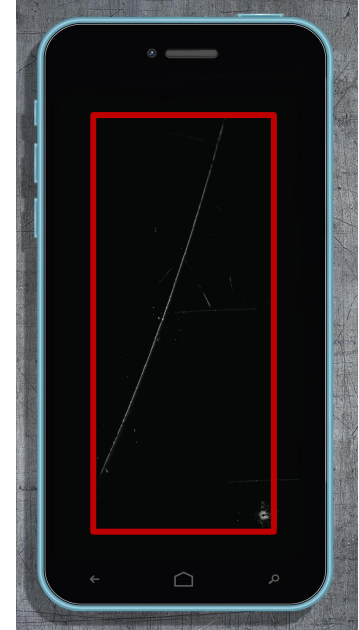
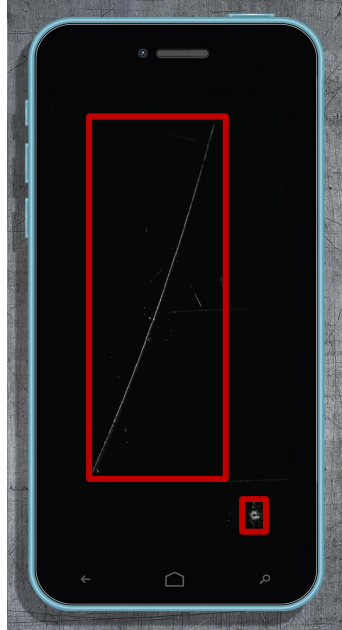
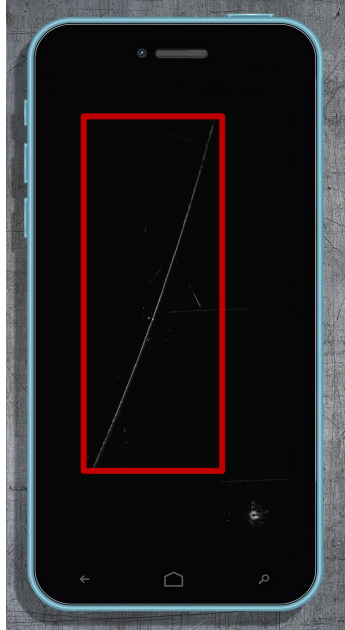
Why is data definition hard?

Iguana detection example

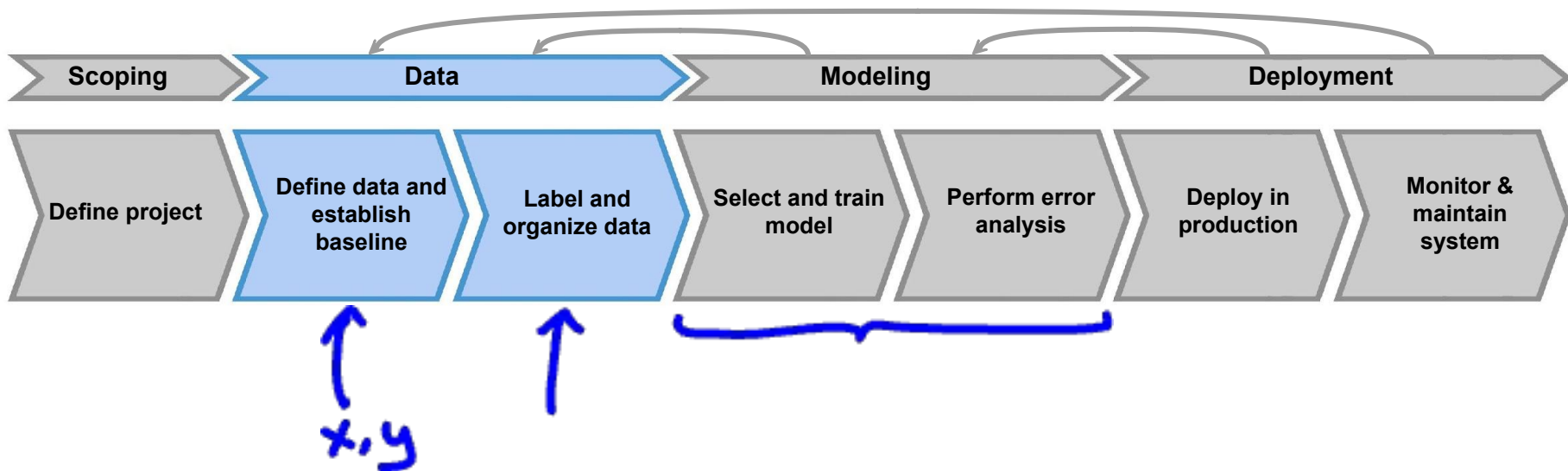


Labeling instructions: "Use bounding boxes to indicate the position of iguanas"

Phone defect detection



Data stage





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Define data and establish baseline

More label ambiguity examples

Speech recognition example



"Um, nearest gas station"

"Umm, nearest gas station"

"Nearest gas station [unintelligible]"

User ID merge example

	Job Board (website)
Email	nova@deeplearning.ai
First Name	Nova
Last Name	Ng
Address	1234 Jane Way
State	CA
Zip	94304

- is it a bot/crawler account?
- fraudulent transaction?
- looking for job?

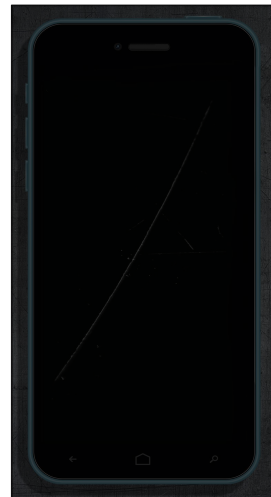


$\left\{ \begin{array}{l} 1 \text{ if same} \\ 0 \text{ if different} \end{array} \right.$



Data definition questions

- What is the input x ?
 - Lightning? Contrast? Resolution?
 - What features need to be included?
- What is the target label y ?
 - How can we ensure labelers give consistent labels?





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Define data and establish baseline

Major types of data problems

Major types of data problems

Unstructured

Structured

Small data

$\leq 10,000$

Clean labels are critical.

Big data

$> 10,000$

Emphasis on data process.

Humans can label data.

Harder to obtain more data.

Data augmentation.

Unstructured vs. structured data

Unstructured data

- May or may not have huge collection of unlabeled examples x .
- Humans can label more data.
- Data augmentation more likely to be helpful.

Structured data

- May be more difficult to obtain more data.
- Human labeling may not be possible (with some exceptions).

Small data vs. big data

Small data

- Clean labels are critical.
- Can manually look through dataset and fix labels.
- Can get all the labelers to talk to each other.

Big data

- Emphasis data process.

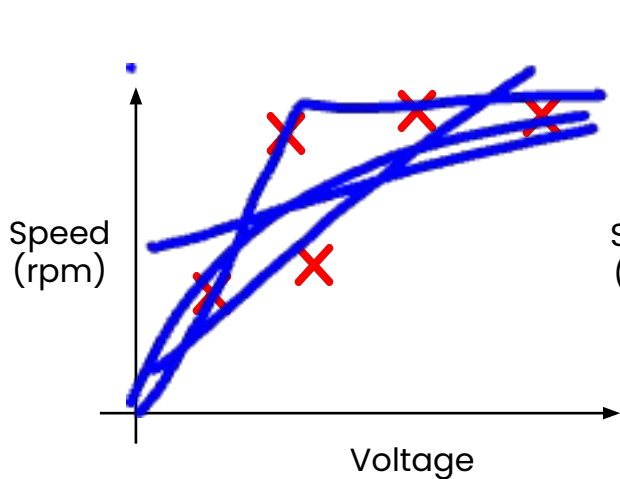


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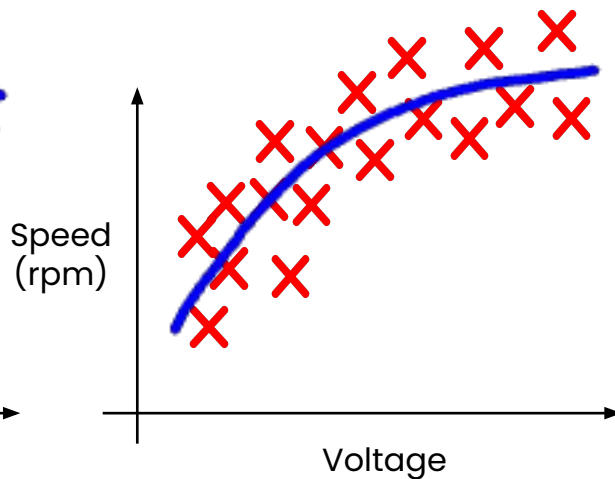
Define data and establish baseline

Small data and label consistency

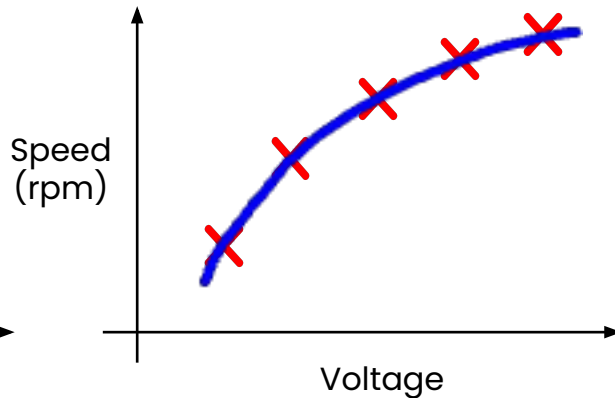
Why label consistency is important



- Small data
- Noisy labels

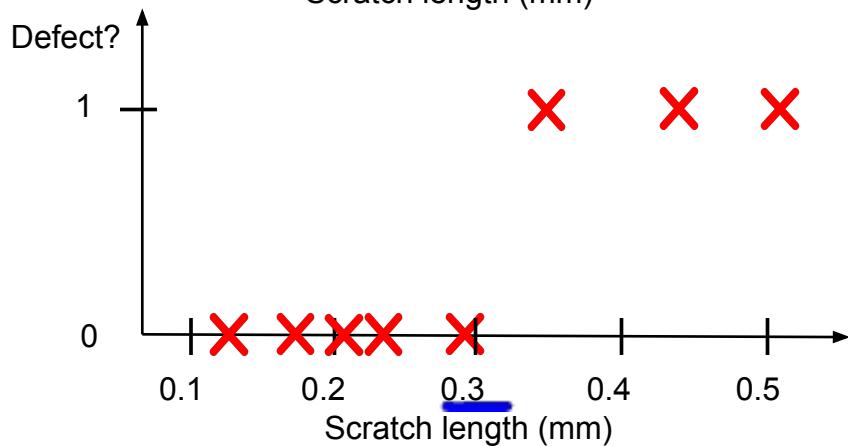
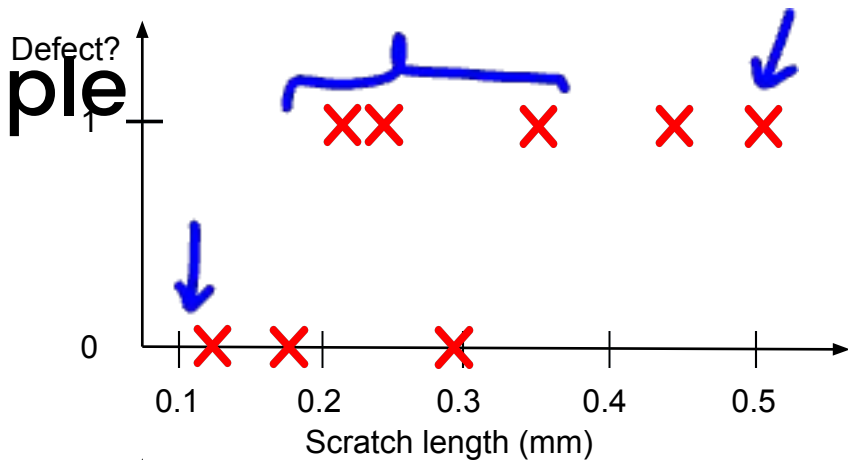
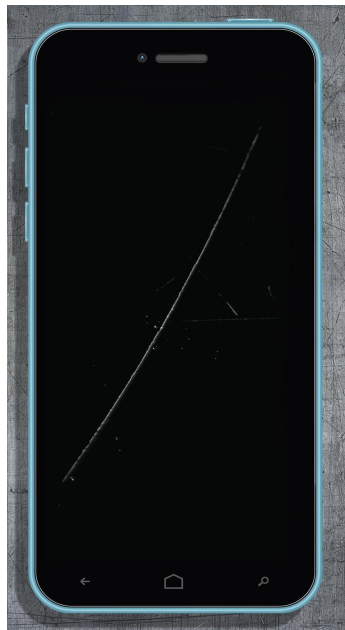


- Big data
- Noisy labels



- Small data
- Clean (consistent) labels

Phone defect example



Big data problems can have small data challenges too

Problems with a large dataset but where there's a long tail of rare events in the input will have small data challenges too.

- Web search
- Self-driving cars ←
- Product recommendation systems ←



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Define data and establish baseline

Improving label consistency

Improving label consistency

- Have multiple labelers label same example.
- When there is disagreement, have MLE, subject matter expert (SME) and/or labelers discuss definition of y to reach agreement.
- If labelers believe that x doesn't contain enough information, consider changing x .
- Iterate until it is hard to significantly increase agreement.

Examples

- Standardize labels

"Um, nearest gas station"

"Umm, nearest gas station"

"Nearest gas station [unintelligible]"



"Um, nearest gas station"

- Merge classes



Deep scratch



Shallow scratch



Scratch

Have a class/label to capture uncertainty

- Defect: 0 or 1



Alternative: 0, Borderline, 1

- Unintelligible audio



“nearest go”

“nearest grocery”

“nearest [unintelligible]”

Small data vs. big data (unstructured data)

Small data

- Usually small number of labelers.
- Can ask labelers to discuss specific labels.

Big data

- Get to consistent definition with a small group.
- Then send labeling instructions to labelers.
- Can consider having multiple labelers label every example and using voting or consensus labels to increase accuracy.





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Define data and establish baseline

**Human level
performance (HLP)**

Why measure HLP?

→ Estimate Bayes error / irreducible error to help with error analysis and prioritization.

Ground
Truth Label

1

1

1

0

0

0

Human?

✓

✗

✓

✓

✓

✗

99%

667% accuracy

Other uses of HLP

- In academia, establish and beat a respectable benchmark to support publication.
- Business or product owner asks for 99% accuracy. HLP helps establish a more reasonable target.
- “Prove” the ML system is superior to humans doing the job and thus the business or product owner should adopt it.

X ← Use with caution

The problem with beating HLP as a “proof” of ML “superiority”

“Um... nearest gas station” ← 70% of labels

“Um, nearest gas station” ← 30%

Two random labelers agree:

$$0.7^2 + 0.3^2 = 0.58$$

ML agrees with humans:

$$\underline{0.70} \leftarrow \begin{matrix} 0.58 \\ + 12\% \end{matrix}$$

The 12% better performance is not important for anything! This can also mask more significant errors ML may be making.



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Define data and establish baseline

Raising HLP

Raising HLP

When the ground truth label is externally defined, HLP gives an estimate for Bayes error / irreducible error. *Very large*

But often ground truth is just another human label.

Ground Truth Label	Inspector
1	1
1 0	0
1	1
0	0
0	0
0	1 0
<i>0.333</i>	<i>↑</i>

66.7%
↓
100%

Raising HLP

- When the label y comes from a human label, $HLP \ll 100\%$ may indicate ambiguous labeling instructions. *Um, Um...*
- Improving label consistency will raise HLP.
- This makes it harder for ML to beat HLP. But the more consistent labels will raise ML performance, which is ultimately likely to benefit the actual application performance.

HLP on structured data

Structured data problems are less likely to involve human labelers, thus HLP is less frequently used.

Some exceptions:

- User ID merging: Same person?
- Based on network traffic, is the computer hacked?
- Is the transaction fraudulent?
- Spam account? Bot?
- From GPS, what is the mode of transportation – on foot, bike, car, bus?

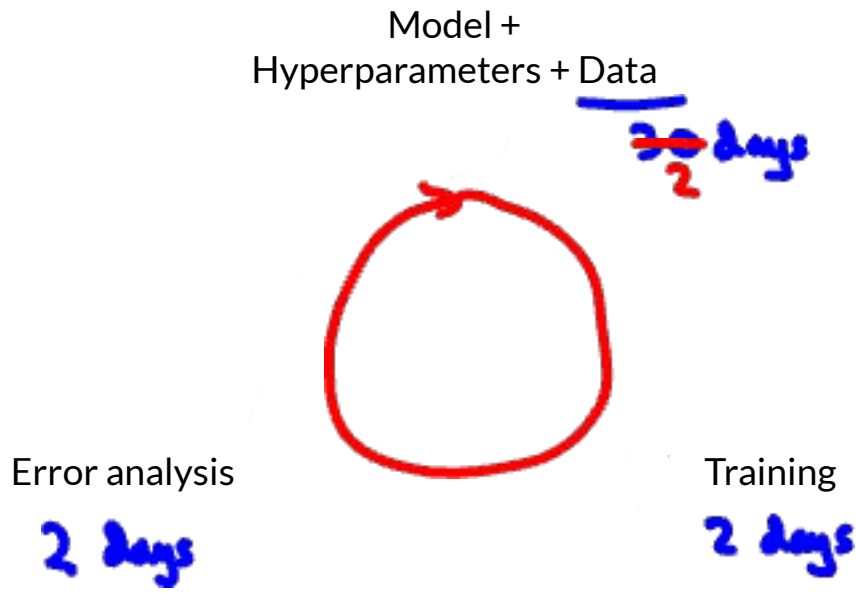


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Label and organize data

Obtaining data

How long should you spend obtaining data?



- Get into this iteration loop as quickly as possible.
- Instead of asking: How long it would take to obtain m examples?
Ask: How much data can we obtain in k days.
- Exception: If you have worked on the problem before and from experience you know you need m examples.

Inventory data

Brainstorm list of data sources ( speech recognition)

Source	Amount	Cost
Owned	100h	\$0
Crowdsourced – Reading	1000h	\$10000
Pay for labels	100h	\$6000
Purchase data	1000h	\$10000



Other factors: Data quality, privacy, regulatory constraints

Labeling data

- Options: In-house vs. outsourced vs. crowdsourced
- Having MLEs label data is expensive. But doing this for just a few days is usually fine.
- Who is qualified to label?
 -  Speech recognition – any reasonably fluent speaker
 -  Factory inspection, medical image diagnosis – SME (subject matter expert)
 -  Recommender systems – maybe impossible to label well
- Don't increase data by more than 10x at a time



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Label and organize data

Data pipeline

Data pipeline example

	Job Board (website)	Resume chat (app)
Email	nova@deeplearning.ai	nova@chatapp.com
First Name	Nova	• Nova
Last Name	Ng	Ng
Address	1234 Jane Way	?
State	CA	?
Zip	94304	94304

x = user info

y = looking for job

Raw data



Data cleaning

spam
cleanup

→ user ID merge

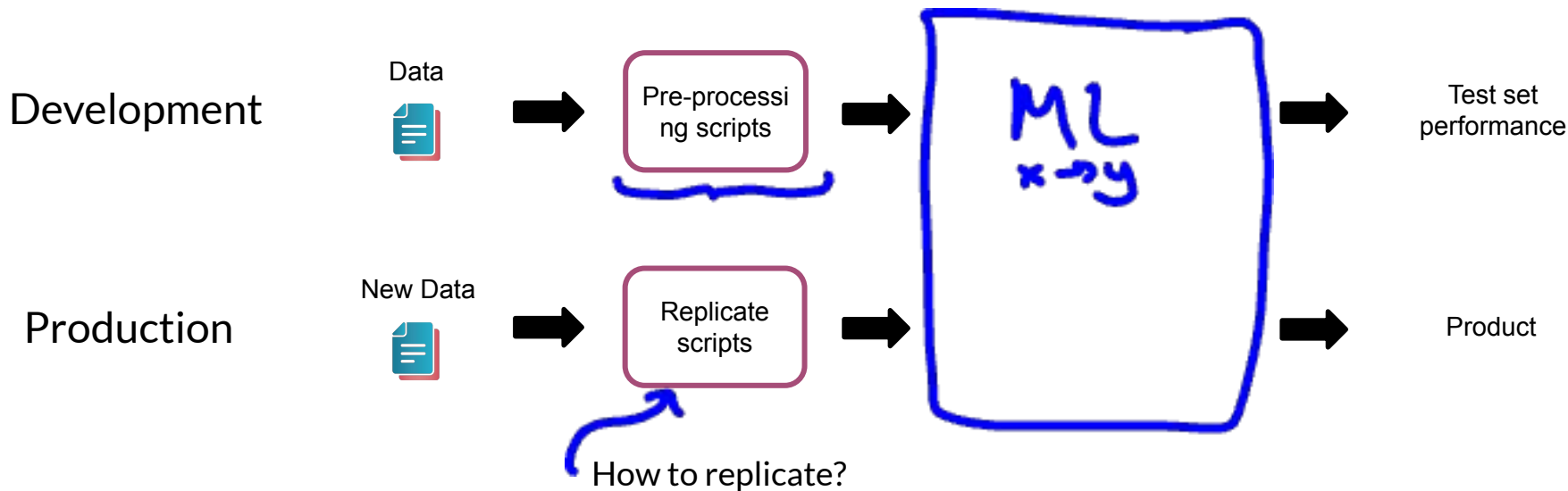
scripts



ML

to predict y

Data pipeline example



POC and Production phases

POC (proof-of-concept):

- Goal is to decide if the application is workable and worth deploying.
- Focus on getting the prototype to work!
- It's ok if data pre-processing is manual. But take extensive notes/comments.

Production phase:

- After project utility is established, use more sophisticated tools to make sure the data pipeline is replicable.
- E.g., TensorFlow Transform, Apache Beam, Airflow,....



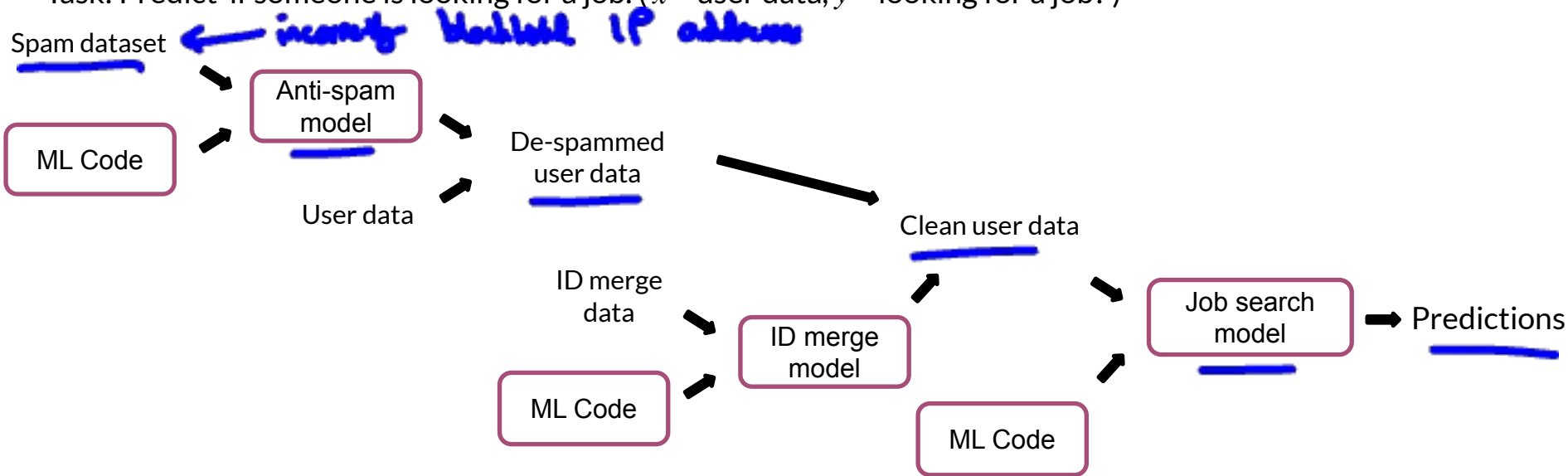
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Label and organize data

**Meta-data, data
provenance and lineage**

Data pipeline example

Task: Predict if someone is looking for a job. (x = user data, y = looking for a job?)



Keep track of data provenance and lineage

where it comes from → sequence of steps

Meta-data

Examples:



Manufacturing visual inspection: Time, factory, line #, camera settings, phone model,
inspector ID,....



Speech recognition: Device type, labeler ID, VAD model ID,....

line 17, today 2

xy

Useful for:

- Error analysis. Spotting unexpected effects.
- Keeping track of data provenance.



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Label and organize data

**Balanced
train/dev/test
splits**

Balanced train/dev/test splits in small data problems



Visual inspection example: 100 examples, 30 positive (defective)

Train/dev/test: 60% / 20% / 20%

Random split: 21 / 2 / 7 positive example
 35% 10% 35%

Want: 18 / 6 / 6 } balanced split
 30% / 30% / 30%

No need to worry about this with large datasets – a random split will be representative.



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C1W3 Slides (Optional)

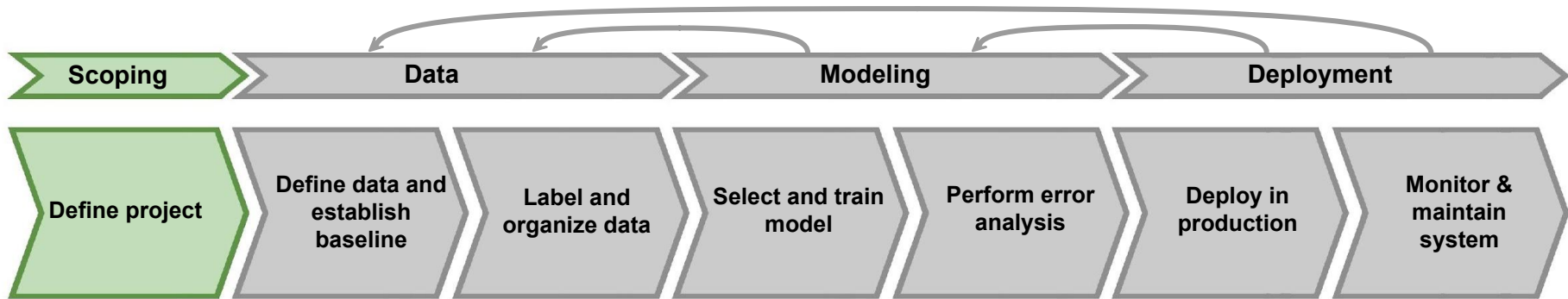
Scoping



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Scoping (optional)

What is scoping?



Scoping example:  Ecommerce retailer looking to increase sales

- Better recommender system
- Better search
- Improve catalog data
- Inventory management
- Price optimization

Questions:

- What projects should we work on?
- What are the metrics for success?
- What are the resources (data, time, people) needed?

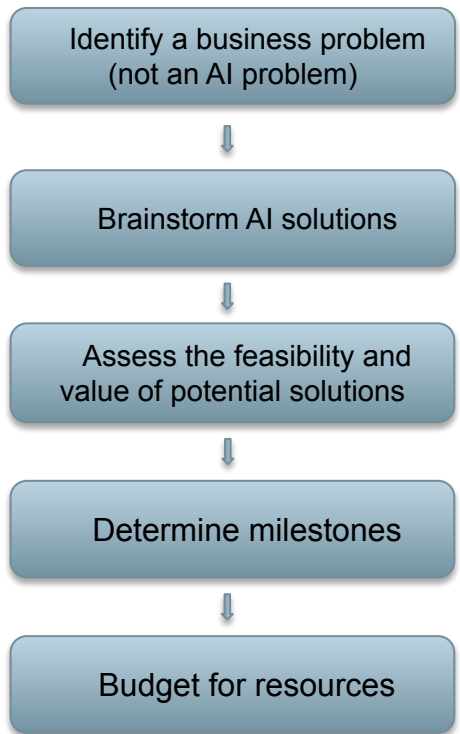


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Scoping (optional)

Scoping process

Scoping process



What are the top 3 things you wish were working better?

- Increase conversion
- Reduce inventory
- Increase margin (profit per item)

Separating problem identification from solution

Problem	Solution
Increase conversion	Search, recommendations
Reduce inventory	Demand prediction, marketing
Increase margin (profit per item)	Optimizing what to sell (e.g., merchandising), recommend bundles
What to achieve	How to achieve



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Scoping (optional)

**Diligence on feasibility and
value**

Feasibility: Is this project technically feasible?

Use external benchmark (literature, other company, competitor)

	Unstructured (e.g., speech, images)	Structured (e.g., transactions, records)
New	<u>HLP</u>	<u>Predictive features available?</u>
Existing	<u>HLP</u> <u>History of project</u>	<u>New predictive features?</u> <u>History of project</u>

HLP: Can a human, given the same data, perform the task?

Why use HLP to benchmark?

People are very good on unstructured data tasks

Criteria: Can a human, given the same data, perform the task?



Do we have features that are predictive?

x

.

y



Given past purchases, predict future purchases



Given weather, predict shopping mall foot traffic



Given DNA info, predict heart disease



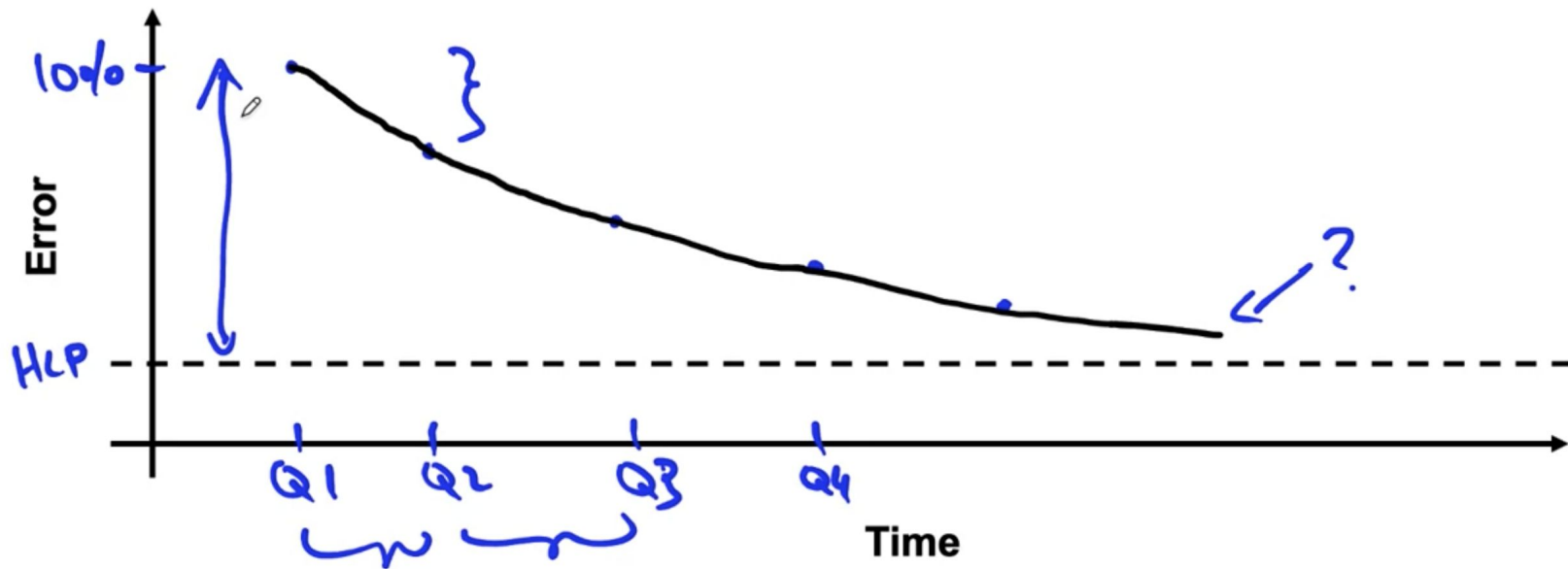
Given social media chatter, predict demand for a clothing style



Given history of a stock's price, predict future price of that stock



History of project



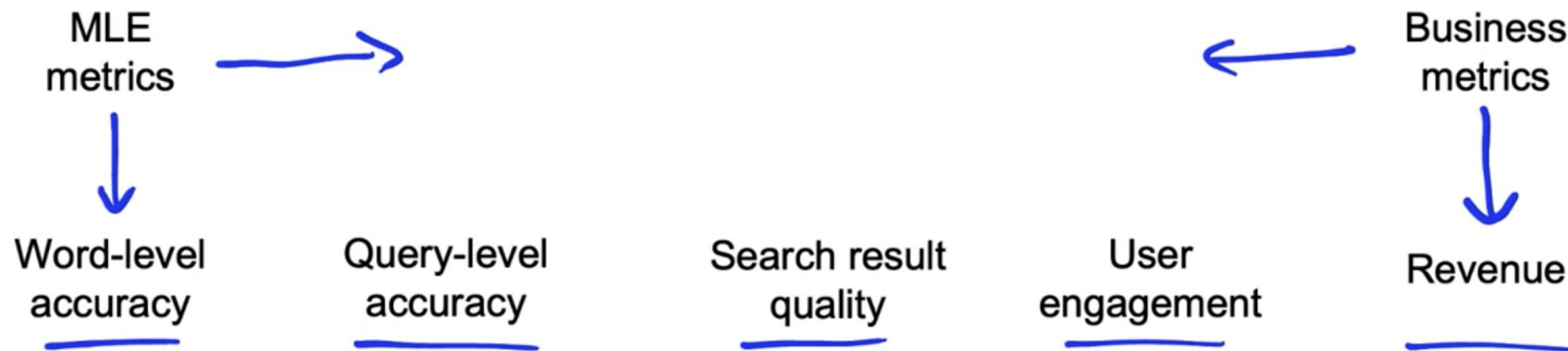


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Scoping (optional)

**Diligence on
value**

Diligence on value



Have technical and business teams try to agree on metrics that both are comfortable with.

Fermi estimates

Ethical considerations

- Is this project creating net positive societal value?
- Is this project reasonably fair and free from bias?
- Have any ethical concerns been openly aired and debated?



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Scoping (optional)

Milestones and resourcing

Milestones

Key specifications:

- ML metrics (accuracy, precision/recall, etc.)
- Software metrics (latency, throughput, etc. given compute resources)
- Business metrics (revenue, etc.)
- Resources needed (data, personnel, help from other teams)

Timeline

If unsure, consider benchmarking to other projects, or building a POC (Proof of Concept) first.



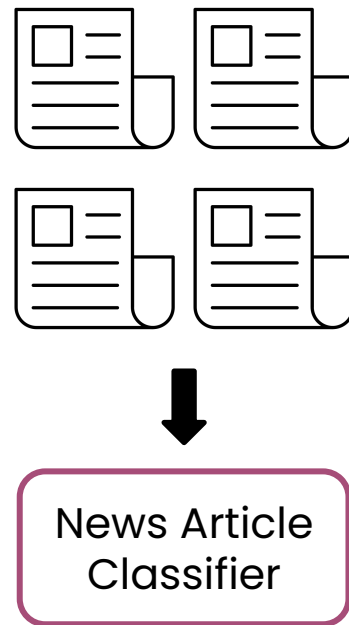
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Final project

Final project overview

Project overview

- Classify news articles
- Start from an existing prototype
- Iteratively improve the performance of the system



Techniques

- Establish a baseline
- Balanced train/dev/test split
- Error analysis
- Track experiments
- Deploy using Tensorflow Serving



DeepLearning.AI Community
community.deeplearning.ai/c/ai-projects/